property of the world (like the mass of the electron or the existence of three families of particles) for which there is currently no explanation. In this chapter we will discuss how far string theorists have gone toward reaching these goals.

Ironically, we will see that although string theory has the potential to be the most predictive theory that physicists have ever studied—a theory that has the capacity to explain the most fundamental of nature's properties—physicists have not as yet been able to make predictions with the precision necessary to confront experimental data. Like a child who receives his or her dream gift for Christmas but can't quite get it to work because a few pages of the instructions are missing, today's physicists are in possession of what may well be the Holy Grail of modern science, but they can't unleash its full predictive power until they succeed in writing the full instruction manual. Nevertheless, as we discuss in this chapter, with a bit of luck, one central feature of string theory could receive experimental verification within the next decade. And with a good deal more luck, indirect fingerprints of the theory could be confirmed at any moment.

Crossfire

Is string theory right? We don't know. If you share the belief that the laws of physics should not be fragmented into those that govern the large and those that govern the small, and if you also believe that we should not rest until we have a theory whose range of applicability is limitless, string theory is the only game in town. You might well argue, though, that this highlights only physicists' lack of imagination rather than some fundamental uniqueness of string theory. Perhaps. You might further argue that, like the man searching for his lost keys solely under a street light, physicists are huddled around string theory merely because the vagaries of scientific history have shed one random ray of insight in this direction. Maybe. And, if you're either relatively conservative or fond of playing devil's advocate, you might even say that physicists have no business wasting time on a theory that postulates a new feature of nature some hundred million billion times smaller than anything we can directly probe experimentally.

If you voiced these complaints in the 1980s when string theory first made its splash, you would have been joined by some of the most respected physicists of our age. For instance, in the mid-1980s Nobel